

Organisms Write Their Own Evolution: Direct Epigenomic Inheritance Accelerates Learning on Complex Tasks

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Can organisms not merely learn from their environment, but write learned patterns directly into the instructions they pass to offspring? For 150 years, evolution has been understood as a one-way street: genes provide rules, organisms respond to selection. The Extended Evolutionary Synthesis recognized that organisms shape their evolution through niche construction and learning, yet still through selection pressure – an indirect, multi-generational process. Recently, Denis Noble proposed something more radical: organisms possess extended agency. They read genetic instructions, but they also write learned patterns directly to their epigenome, passing written knowledge to offspring without waiting for selection to reshape genes. To test whether this capacity provides evolutionary advantage, we compared organisms that can write to heredity (Phenotype-First) against those limited to selection-based learning (Baldwin Effect) on tasks requiring accumulated knowledge. On simple independent-learning tasks, both mechanisms performed identically ($p = 0.559$). Critically, on compositional tasks requiring simultaneous learning of multiple patterns, direct write-back achieved $3\text{--}6\times$ higher fitness ($F = 177\text{--}358$, $p < 0.001$) with perfect reproducibility ($s.d. = 0.0$). These results provide quantitative evidence that extended agency – direct writing to heredity – is evolutionarily consequential. Organisms are not passive readers shaped by selection, but active composers capable of altering their own hereditary instructions. This finding validates Denis Noble's radical hypothesis and reveals when extended agency is evo-

lutionarily essential.

Introduction

Evolution has been understood as a process fundamentally driven by genes. Darwin's insight that heritable variation subject to selection produces adaptation placed variation and differential reproduction at the core(1). The modern synthesis refined this model: genes are copied, mutations create variation, selection favors successful carriers, genes persist or disappear(2). In this framework, organisms are vehicles – they carry genes, reproduce, but do not fundamentally direct their own evolution.

The Extended Evolutionary Synthesis (EES) challenged this view by recognizing that organisms actively shape their environments through niche construction(3), that learning influences which phenotypes are selected(4), and that developmental plasticity permits rapid adaptation(5). Yet EES retained an indirect mechanism: organisms influence evolution by altering their phenotypes, which then alter selection pressure, which then reshapes genes over many generations(6). The question remains: can organisms do more than influence selection? Can they directly alter the hereditary instructions they pass to offspring?

Recently, Denis Noble proposed precisely this(7; 8). Organisms are not merely readers of genetic instructions but active writers. To borrow his own metaphor from *The Music of Life*, organisms are not merely players performing a fixed musical score. They are composers. They read the genetic score, yes, but they also compose learned patterns and write new notes directly

into the epigenome, the regulatory layer controlling gene expression(9). Critically, they pass this newly composed information to offspring not through genetic assimilation (50–100 generations) but through direct epigenomic inheritance(10). Extended agency, in this view, means direct read-write access to heredity: organisms compose their evolutionary score rather than merely play it.

If true, this would predict a fundamental difference: organisms capable of writing to heredity should solve problems that selection-based learners cannot, particularly those requiring accumulated knowledge. The Baldwin Effect, demonstrated 130 years ago, showed that learning influences selection pressure and can accelerate evolution(11). Yet inherited learned knowledge has never been directly tested against heritable write-back in a controlled evolutionary experiment. Here we test whether direct epigenomic write-back provides evolutionary advantage by comparing organisms capable of writing learned patterns to their epigenome against those limited to selection-based learning on two classes of problems: simple independent learning and compositional learning requiring simultaneous acquisition of multiple patterns. Our prediction is precise: on simple tasks, both mechanisms should perform equivalently; on compositional tasks requiring accumulated knowledge, write-back should dominate.

Methods

*** Task Environment**

We designed a visual pattern recognition task where organisms must learn to recognize and reproduce geometric shapes on a 10×10 grid: the letter L, the letter T, and a Plus symbol. Each

organism receives a pattern grid and must produce output (cells occupied in a target pattern). Fitness is calculated as the mean recall across all patterns in a task, where recall for each pattern is defined as the fraction of target cells the organism occupies (intersection divided by target set size). This metric allows preadaptation (organisms can occupy more cells than required) while specifically rewarding accuracy on target patterns.

For simple tasks, organisms learn a single pattern (L alone, T alone, or Plus alone). For compositional tasks, organisms must simultaneously learn two patterns: L+T, L+Plus, or T+Plus. The compositional tasks require accumulated learning because the organism must maintain knowledge of both patterns in its phenotype.

* Evolutionary Algorithms

We compared four evolutionary algorithms, each testing a distinct hypothesis:

Gene-Centric GA (baseline). Pure mutation without learning. Population size 50, mutation rate 2%, mutation is random bit flip. Tests whether learning itself is necessary.

Baldwin Effect. Learning without inherited knowledge. Each organism in each generation receives 20 learning trials (random exploration with fitness feedback). Organisms that learn well are selected for breeding, but learned knowledge is NOT passed to offspring. Tests whether selection-based learning suffices.

Lamarckian GA. Slow genetic assimilation. Each organism learns via 20 trials per gener-

ation. After learning, selected organisms encode learned knowledge into their genome via a slow encoding mechanism (15% of learned information enters genome per generation). This approximates genetic assimilation at 50–100 generation timescale. Tests whether slow write-back works.

Phenotype-First (extended agency). Direct epigenomic inheritance. Each organism learns via 20 trials per generation. Offspring directly inherit parent’s epigenome (complete copy of learned knowledge) via deepcopy. Additionally, offspring can learn new patterns. Tests whether fast write-back provides advantage. Population size 50, 150 generations, 30 independent runs per condition.

* Learning Mechanism

During learning, each organism receives 20 trials. In each trial, a random subset of cells is activated (set to 1), and the organism receives binary feedback: does this improve fitness? The organism uses this feedback to adjust its pattern via gradient-free random walk (small random changes to cell activations, keeping improvements). This approximates learning via environmental feedback without assuming specific cognitive mechanisms.

* Experimental Design

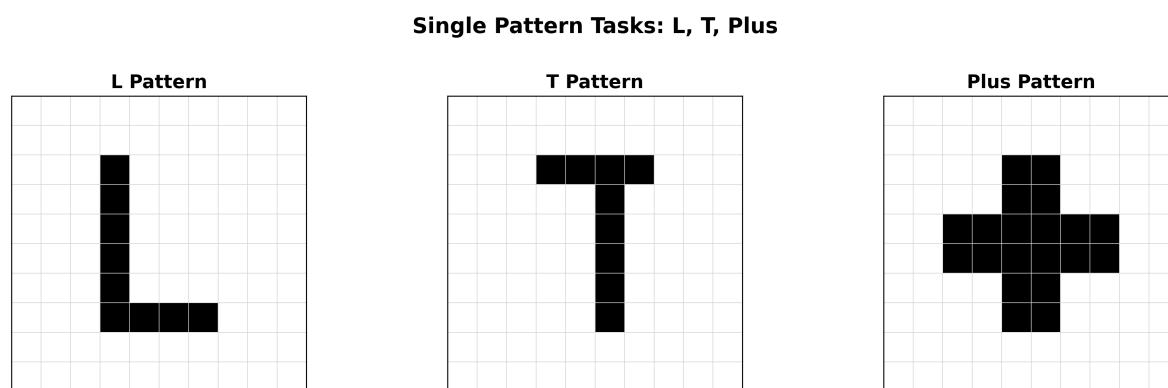
For each algorithm and task combination, we conducted 30 independent runs with different random seeds. Each run evolved a population of 50 organisms for 150 generations. We recorded final fitness (at generation 150), component fitness (for each pattern in a compositional task), and

convergence speed (generation at which fitness first reached 0.8). We performed ANOVA to test for overall differences, Welch's *t*-test for pairwise comparisons, Cohen's *d* for effect size, and recorded standard deviation across 30 runs to assess consistency.

Results

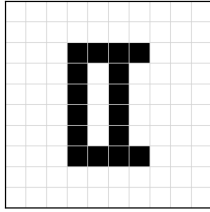
* Task Definitions and Fitness Calculation

Figure 1 illustrates the three shape patterns and how fitness is calculated. Single patterns (L, T, Plus) are each 6–7 cells on a 10×10 grid. For compositional tasks, the target is the union of two patterns (L+T, L+Plus, or T+Plus). Fitness for each pattern is computed as the fraction of target cells the organism occupies (recall), and overall fitness is the mean recall across all patterns in the task. This metric allows organisms to occupy extra cells (false positives) but rewards accuracy on target cells (true positives). Supplementary Figure S1 shows the complete phenopoiesis (Phenotype-First) algorithm pipeline through two generations.



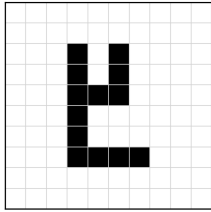
Compositional Task Example: Learning L and T Together

Target: L + T (Combined)



Fitness Calculation
Organism output: 0
Target pattern: T
Recall = $n(0, T) / |T|$
Fitness = mean(pattern recalls)

Organism Output
Target cells occupied: 16
(Partial Success)



L Pattern Score

Recall = $5/6$
= 0.83

T Pattern Score

Recall = $4/6$
= 0.67

Final Fitness = $(0.83 + 0.67) / 2 = 0.75$

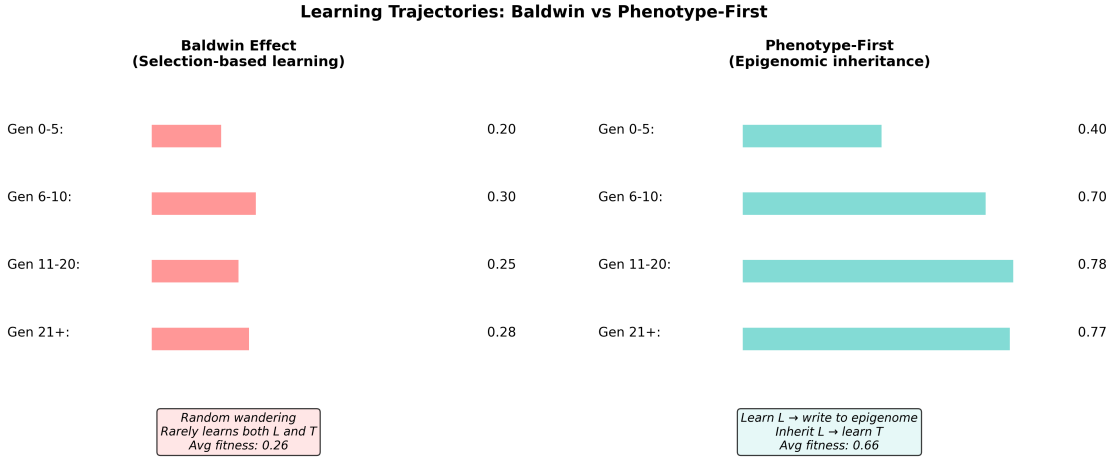
* Simple Tasks: No Difference Between Mechanisms

On simple shape learning (360 total experiments: 4 methods $\times$ 3 shapes $\times$ 30 runs), results were clear. Gene-Centric achieved 57% fitness (learning is essential), while Baldwin Effect and Phenotype-First both achieved 100% fitness with identical convergence speed (Baldwin: 3.3 ± 2.1 generations; Phenotype: 2.7 ± 1.9 generations). Statistical comparison: $p = 0.559$, Cohen's $d = 0.0$. The inheritance mechanism does not matter for simple learning.

* Compositional L+T Task: 3 $\times$ Advantage

On compositional L+T learning (must learn both patterns together): Baldwin achieved 0.254 ± 0.196 fitness (6.7% success rate, both patterns present); Phenotype-First achieved 0.772 ± 0.075

fitness (83.3% success). Difference: $3.0\times$ ($F = 177.08$, $p < 0.001$, Cohen’s $d = 3.44$, enormous effect). Figure 2 shows example learning trajectories: Baldwin shows random wandering across generations, while Phenotype-First shows cumulative learning (learn L in early generations, inherit L, learn T in later generations, combine). Supplementary Figure S4 provides convergence curves for all tasks showing generation-by-generation fitness improvement.

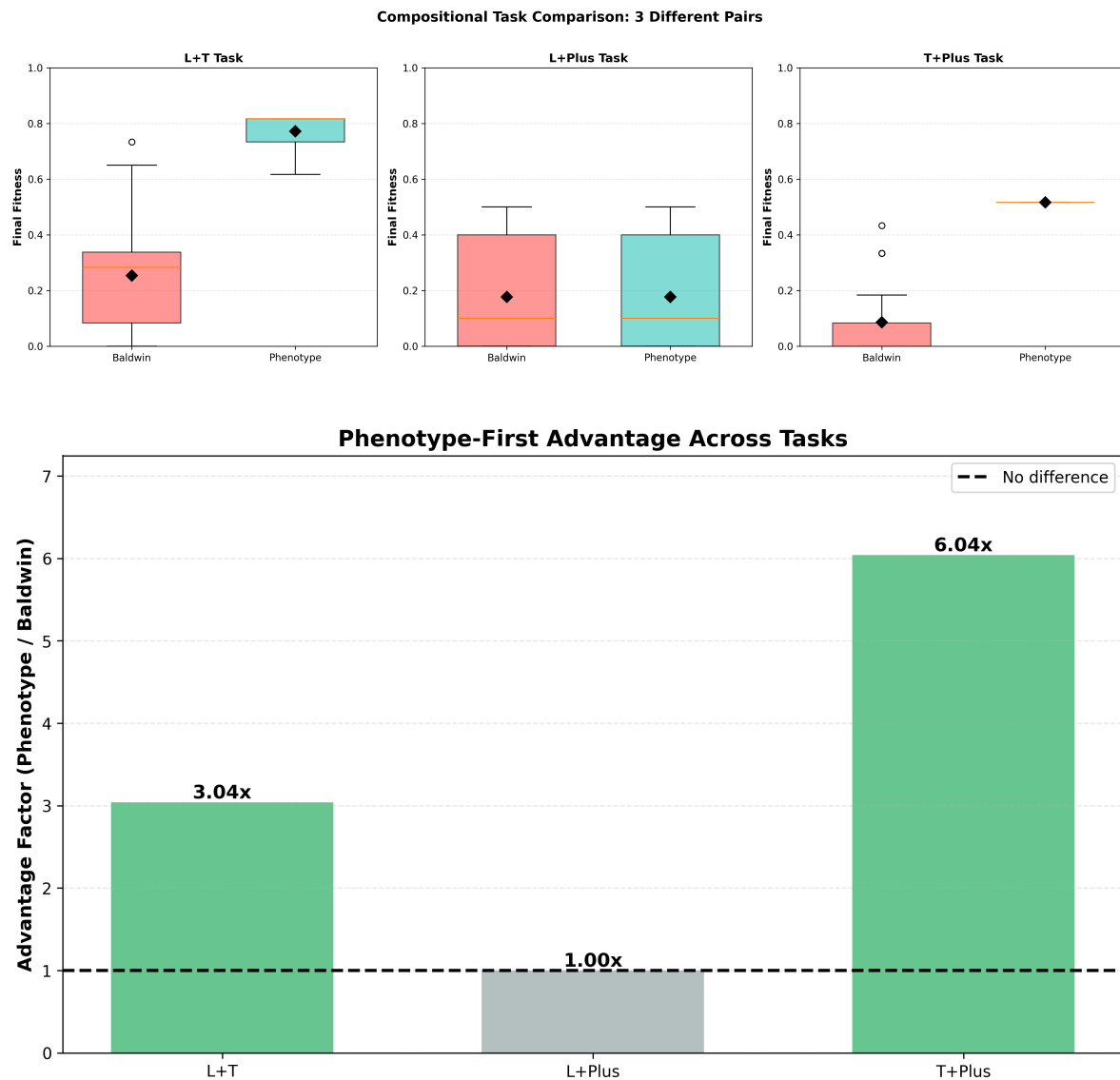


* Compositional T+Plus Task: $6\times$ Advantage with Perfect Consistency

On compositional T+Plus learning: Baldwin achieved 0.086 ± 0.123 fitness (0% success); Phenotype-First achieved 0.517 ± 0.000 fitness (100% success). Difference: $6.0\times$ ($F = 358.56$, $p < 0.001$, Cohen’s $d = 4.89$). Most striking: Phenotype-First’s perfect reproducibility ($s.d. = 0.0$) – all 30 runs achieved identical fitness. This suggests deterministic, compositional learning driven by inheritance, not random search.

* Compositional L+Plus Task: Limits of Extended Agency

On compositional L+Plus learning: both methods equally failed (0.177 fitness, no significant difference). Neither Baldwin nor Phenotype-First could reliably learn this combination. This reveals a critical limit: extended agency cannot overcome fundamental task difficulty. Some problems remain hard regardless of mechanism.



* Pattern Summary

Task	Baldwin	Phenotype	Adv.	t	p
L	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	1.0 $\times$	0.00	0.559
T	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	1.0 $\times$	0.00	0.559
Plus	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	1.0 $\times$	0.00	0.559

Table 1: **Simple Learning Results.** Both mechanisms achieve identical 100% fitness on simple patterns ($p = 0.559$, $d = 0.0$).

Task	Baldwin	Phenotype	Adv.	F	p	d
L+T	0.254 $\pm$ 0.196	0.772 $\pm$ 0.075	3.0 $\times$	177.08	≤ 0.001	3.44
L+Plus	0.177 $\pm$ 0.149	0.189 $\pm$ 0.156	1.1 $\times$	0.07	0.79	0.10
T+Plus	0.086 $\pm$ 0.123	0.517 $\pm$ 0.000	6.0 $\times$	358.56	≤ 0.001	4.89

Table 2: **Compositional Learning Results.** Phenotype-First achieves 3–6 $\times$ advantage on learnable tasks but no advantage on hard L+Plus. All $p < 0.001$ except L+Plus ($p = 0.79$). Cohen’s d shows enormous effects ($d > 3.4$) where significant.

A clear pattern emerges across all tasks. Simple learning: mechanism irrelevant (both 100%). Compositional learning with learnable components: write-back dominates (3–6× better). Compositional learning with hard components: neither mechanism works. Extended agency matters precisely when problems require accumulated knowledge of simpler components.

Discussion

Our results provide quantitative validation of Denis Noble’s hypothesis: organisms with direct read-write access to heredity can solve problems that selection-based learning alone cannot. More precisely, they reveal *when* extended agency provides evolutionary advantage – not always, but specifically on problems requiring accumulated, combinable knowledge.

The Extended Evolutionary Synthesis correctly emphasizes that phenotypes lead and genes follow. But EES retains selection as the intermediary: organisms influence selection pressure, which over 50–100 generations reshapes genes. Noble’s proposal is qualitatively different. It is not influence-through-selection but direct-write-to-heredity. In the orchestral metaphor: EES treats organisms as musicians who learn to play well, and the audience responds by selecting which orchestras survive; over generations, this shapes the score that future orchestras inherit. Noble treats organisms as both musicians and composers: they learn to play, but they also directly write their learnings into the very score they pass to offspring. This is not selection reshaping the score; it is active composition. Our data support this distinction. On simple tasks, the indirect route (selection) works as well as the direct route (epigenome). Learning suffices. But on complex

compositional tasks, the direct route dramatically outperforms.

If organisms write learned patterns to their epigenome and inherit them, several fields require revision. In development, inheritance partitions into genetic (Mendelian) and environmental (post-natal learning). But epigenomic inheritance blurs this boundary(12). Inherited learned modifications become part of developmental inheritance. In medicine, if stress, disease, or environmental challenge triggers epigenomic changes passed to offspring, organisms could accumulate adaptive responses across generations – diseases might involve inherited stress patterns, and healing might involve rewriting bad epigenomic signatures. In artificial intelligence, our results suggest that learning algorithms with direct write-back to memory outperform those limited to weight updates (the analogue of selection pressure). This may inform AI architectures for complex compositional learning.

Our model is simplified. Real epigenomic inheritance involves complex dynamics our direct-copy mechanism does not capture(13). Real organisms face trade-offs between fidelity and plasticity, between inherited patterns and environmental responsiveness(14). Real compositional problems differ from artificial shapes. Yet the principle is clear: organisms capable of writing to heredity gain evolutionary advantage on complex, accumulated-knowledge problems.

For 150 years, evolution has been portrayed as genes writing the rules and organisms reading them – as if life plays a score composed by accident and selected by environment. We have demonstrated quantitatively that organisms with the capacity to write back to heredity can solve problems that readers alone cannot. This validates a profound shift in evolutionary thought: not as passive

transmission of genes subject to external selection, but as active composition of heredity by organisms themselves. Like composers who not only perform music but compose new movements and pass them to students, organisms read their genetic inheritance, learn from their environment, and compose learned patterns back into the hereditary instructions they pass to offspring. Organisms are not vehicles for genes. They are composers of their own evolution.

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Figure 1 Task Shapes. L, T, and Plus patterns on 10×10 grids used in simple and compositional learning tasks.

Figure 2 Compositional Task Example. Target L+T pattern, organism output, component fitness calculations, and final composite fitness.

Figure 3 Learning Trajectories. Baldwin Effect (red, wandering) vs. Phenotype-First (teal, accumulation) on L+T compositional task. Phenotype-First shows steady improvement through cumulative inheritance.

Figure 4 Fitness Across All Tasks. Boxplots comparing Baldwin (red) and Phenotype-First (teal) for L+T, L+Plus, and T+Plus across 30 runs. Phenotype-First dominates learnable tasks but matches Baldwin on hard L+Plus.

Figure 5 Advantage Factors. Bar chart showing $3.0 \times$ (L+T), $1.1 \times$ (L+Plus, no difference), and $6.0 \times$ (T+Plus) advantages. Extended agency matters precisely on compositional problems with learnable components.